REDCap Newborn Follow-up Dashboard

Standard Operating Procedure (SOP) & Technical Reference Manual — Version 3.0

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Target Audience: Data Analysts, Telecallers, District Supervisors, State Coordinators	**Operational Status:** Live Production Release

1. Executive Overview & System Architecture

This Standard Operating Procedure (SOP V3.0) defines the end-to-end data pipeline, column mappings, data cleansing algorithms, 7-step chronological cross-dataset matching criteria, and data validation rules for the **REDCap Newborn Follow-up Dashboard**. The system aggregates birth notification, registration, and telephonic follow-up data (60,000+ records) from REDCap alongside state ANMOL delivery reports and institutional SNCU admission datasets, providing real-time KPI monitoring, 1-click line-list filtering, and automated discrepancy reporting.

Stage	Data Source / Component	Description & Purpose
1. Ingestion	REDCap, ANMOL, & SNCU Files	Loads raw REDCap dataset (`NHRP update.xlsx`), ANMOL delivery line-lists, and SNCU admissions using Calamine engine.
2. Standardization	Data Cleansing Modules	Parses multi-format dates, swaps inverted day-month entries, standardizes blocks/villages, and validates 10-digit phones.
3. Spatial Mapping	Location & Staff Directories	Maps Block (Col AF) and Village to Subcentre, attaching ANM, CHO, and Telecaller coordinator metadata.
4. ANMOL Match	4-Tier Matching Engine	Cross-matches ANMOL home and out-of-district births with REDCap records via MPID, phone, and fuzzy name similarity.
5. SNCU Match	7-Step Chronological Engine	Cross-verifies SNCU neonatal deaths (<=28 days) with REDCap across 7 sequential criteria with dual record IDs.
6. Validation	Form 1 & Form 2 Rules	Evaluates 17 Form 1 checks (excluding pending cases in 1.9) and 7 Form 2 checks (evaluating Col XU/AAD/ACM).
7. Standalone Build	HTML Generator (`generate_dashboard.py`)	Bundles compressed JSON payload, Chart.js v4, and interactive UI into a self-contained offline HTML file.

2. Input Data Files & Column Mapping Reference

The dashboard generator integrates multiple heterogeneous datasets. Below are the exact Excel column headers, Excel column numbers, and operational roles used in the current version:

Category	Column No. & Exact Header Name	Variable Description & Usage
Record Key	`Record ID`	Primary key unique to each REDCap registration record.

Block Hierarchy	**Col AF:** `Block / Taluk Name `	Administrative health block used for spatial filtering and match criteria Step 1.
Delivery Place	**Col BY:** `Q43. If health facility, Delivery Place name (Public)23`	Public health facility name used for place filtering and match criteria Step 2.
Telecallers	`Q10. TeleCaller/Data Collector ID` **Col 639/700/761:** `Q10. TeleCaller ID410 / ID471 / ID532`	Assigned telecaller coordinators across registration and follow-up calls 1–3 (23 active coordinators).
Delivery Dates	`Q33. Delivery Date` **Col XO/ZX/ACG:** `Q9. Call Date_Time 409/470/531`	Delivery date and follow-up completion timestamps used for activity reporting.
Follow-up Status	**Col XU/AAD/ACM:** `Q15. Baby Current Status 415 / 476 / 537`	Final follow-up outcome status (Healthy, Dead, Sick, Completed) for Baby 1, 2, and 3.
Mortality Data	`Q51 / Q57 / Q63`: Date of death `Q16. If baby died, Date of death...` `Q18` series: Place of death	Neonatal death dates, place of death (Hospital/Home/Way), and age at death (<= 28 days).

3. 7-Step Chronological SNCU Neonatal Death Matching Engine

To cross-verify hospital-based neonatal mortality (SNCU DH Khargone) with community REDCap records, the pipeline applies a strict chronological 7-step criteria hierarchy. Expired cases with 0 ≤ Age at Death ≤ 28 days (calculated from Date of Birth to Discharge Date) are evaluated:

Step No.	Matching Field / Variable	REDCap Source Field	SNCU Admission Field
Step 1	Block / Taluk Name	**Col AF:** `Block / Taluk Name `	`Block Name`
Step 2	Place of Delivery	**Col BY:** `Q43. Delivery Place name...`	`Place of Delivery`
Step 3	Delivery Date vs Date of Birth	`Q33. Delivery Date`	`Date of Birth`
Step 4	Name of Mother & Father	`Q2. Mother Name` / `Q3. Father Name`	`Mother Name` / `Father/Husband Name`
Step 5	Primary Contact Number	`Q16. Family Mobile Number`	`Contact No.`
Step 6	Date of Death vs Discharge Date	`Q51/Q57/Q63/Q16` (Date of Death)	`Discharge Date` (Expired date)
Step 7	Secondary Mobile Verification	`Q17. Family 2nd Mobile Number`	`Contact No.`

- **Accurately Recorded (≥ 5 Criteria Matched):** Cases matching on 5 or more criteria out of the 7 chronological steps.
- **Recorded with Spelling Mismatch (< 5 Criteria Matched):** Cases established via DOB + Delivery Place with minor spelling differences in names/phone numbers.
- **Not Recorded in REDCap:** Cases where no corresponding delivery record exists in the REDCap follow-up repository.
- **Dual Record IDs:** Side-by-side display of SNCU Admission ID and REDCap Record ID in expandable inspection drawers.

4. Data Validation Quality Engine (Form 1 & Form 2 Rules)

Automated data quality checks evaluate registration completeness, gestational age consistency, and follow-up outcomes across Form 1 (1.1 to 1.17) and Form 2 (2.1 to 2.7):

Code	Validation Rule Name	Evaluation Formula & Operational Definition

1.1	Delivery Date Missing	`Q33. Delivery Date` is NaN or blank.
1.2	Block Name Missing	`Block / Taluk Name ` (Col AF) is NaN, blank, or 'Not Found'.
1.3	Gestational Age Missing	`Q38 (EDD)`, `Q32 (LMP)`, and `Q35/Q36 (Clinical GA)` are all blank.
1.4	Gestational Age Mismatch	Variance among derived GAs > 1.5 weeks (Priority: EDD > LMP > Clinical).
1.5	Delivery Type Missing	`Q39. Delivery type` is NaN or blank.
1.6	Delivery Location Missing	`Q41. Delivery Location` is NaN or blank.
1.7	Number of Babies Delivered Missing	`Q45. Number of baby delivered` is NaN or blank.
1.8	Baby Not Registered after Notification	`Q45` ≥ 1 baby, but `Q47 (Baby-1 Status)` is blank.
1.9	Status of Baby at Birth Mismatched	Stillbirth reported in `Q47`, but baby has completed follow-up status (Healthy/Dead/Sick) or death date (excludes Pending cases).
1.10	Stillbirth GA < 28wks or Weight < 1000g	`is_stillbirth` is True AND (`gestation` < 28 weeks OR `weight` < 1000g).
1.11	Baby Gender Missing	`Q48 / Q54 / Q60 (Sex)` is blank or 'Unknown'.
1.12	Birth Weight Missing / 9999	Birth weight is NaN, 9999, ≤ 0g, or > 8000g.
1.13	Live Birth Gestational Age < 24wks	`is_live_born` is True AND `gestation` < 24 weeks.
1.14	Intergrowth Chart GA vs BW Mismatch	Birthweight >200g outside -4SD to +4SD for GA 24–42 weeks.
1.15	Mobile Number Issues	Both phones missing OR primary phone fails 10-digit numeric check.
1.16	Duplicate Entries	Multiple rows share identical `(Mother Name, Mobile, Delivery Date)`.
1.17	Future Date Issues	`Delivery Date` > current system date.
2.1	Day 29 Status Missing / Due	Live birth ≥ 29 days old without completed status in Col XU/AAD/ACM.
2.2	Date of Death Missing	Baby reported Dead in calls, but `Date of Death` is NaN.
2.3	Place of Death Missing	Neonatal death confirmed, but `Place of Death` is Not Recorded.
2.4	Age at Death Invalid	`age_at_death` < 0 days OR `age_at_death` > 28 days.
2.5	Factors Associated with ND Missing	Neonatal death confirmed, but cause-of-death checkboxes unchecked.
2.6	Call Date Before Delivery Date	Any follow-up `call_date` < `delivery_date`.
2.7	Death Timeline Issues	Call 3 date > death date OR Call 3 status is Healthy/Sick for deceased baby.

5. Interactive UI Controls & Line List Filtering

The web dashboard provides multi-layered filtering controls to support supervisory and field-level follow-up:

1. **Global Filter Bar:** Filter simultaneously by **Month**, **Block**, **Delivery Coordinator (Col 21)**, and **Follow-up Done By (Col 639/700/761)** across all tabs.
2. **1-Click Interactive KPI Cards:** Instant single-click filtering across Home Deliveries (All, Accurately Recorded, Spelling Mismatch, Unmatched), Out-of-District Deliveries, and SNCU Deaths.
3. **Place of Delivery Sub-Filters:** Dedicated delivery location dropdowns in **Neonatal Deaths Registry (Col BY)**, **Khargone SNCU Deaths**, and **Khandwa SNCU Deaths** line-lists.
4. **Inspection Drawers:** Clickable row drawers showing field-by-field verification matrices, health worker contact cards, and chronological comparison summaries.
5. **Excel CSV Export:** One-click filtered data export with automatic file naming for ground-level distribution.

Prepared By: Technical Data Analyst Team	**Approved By:** State Health Program Director
Signature: _____	**Signature:** _____
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